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The Role of Lower-Limb Muscle Volumes in Vertical Jump Performance in NCAA Division 1 Athletes

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Running Title: MUSCLE VOLUMES AND VERTICAL JUMP PERFORMANCE

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ABSTRACT

Introduction/Purpose: This study examined the relationship between lower extremity muscle volumes and countermovement jump (CMJ) performance in 207 (43 female, 164 male) NCAA Division I collegiate basketball, football, hockey, and soccer athletes. The primary aim was to determine which combination of lower extremity muscle volumes was most strongly associated with CMJ performance across a large subset of high-level college athletes. **Methods:** Athletes underwent bilateral MRI scans, and muscle volumes were quantified using machine-learning-based auto-segmentation. CMJ height (cm) and peak power (W/kg) were measured using dual force plates. Backward selection regression analyses determined muscle combinations that best explained variance in jump performance after adjusting for sex and sport. **Results:** Vastus lateralis muscle volume was related to both jump height ($\beta = 0.36$, $p < 0.01$) and peak power ($\beta = 0.41$, $p < 0.01$). In addition to the vastus lateralis, the jump height model retained rectus femoris, and the peak power model retained soleus, gluteus medius, adductor magnus, and biceps femoris long head. The differing muscle candidates in the two models likely reflect the distinct features of each CMJ metric: jump height reflects force over time (impulse, N·s) while peak power reflects the rapid generation of force (N·m/s). Sport-specific differences emerged, with hockey and football athletes displaying greater jump heights and generating greater peak power when compared to basketball players. Male athletes outperformed females in both jump metrics. **Conclusions:** These findings, derived from over 200 NCAA Division I athletes, highlight the crucial role of the quadriceps in CMJ performance in high-level athletes. Strength and conditioning and rehabilitation programs may consider targeting quadriceps hypertrophy to improve vertical jump performance. **Key Words:** COUNTERMOVEMENT JUMP, MAGNETIC RESONANCE IMAGING, VASTUS LATERALIS, QUADRICEPS

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INTRODUCTION

The ability to generate force and power through the lower extremities during athletic tasks such as jumping is a critical determinant of performance (1–3). The countermovement jump (CMJ) is a foundational movement for various sport-specific skills and is widely used by sports medicine and science practitioners as a time-efficient testing method for assessing lower-body power, performance, and readiness for return to play from injury (4–6).

Larger muscle size is known to relate to greater force and power production (7, 8). However, traditional imaging methods for quantifying muscle volumes to assess function are time-intensive, requiring manual segmentation and extensive processing, limiting the scope of investigations to key muscle groups and smaller sample sizes. As a result, prior research examining the relationship between muscle size and jump performance has primarily focused on a select few muscles, particularly the rectus femoris, vastus lateralis, and lateral gastrocnemius as predictors of jump performance (8–10). Further, most studies have strictly investigated the muscle morphology (e.g. anatomical cross-sectional area or thickness) and architecture (e.g., pennation angle, fascicle length, and physiological cross-sectional area) of large, primary muscles, but less is known about the role of other muscles (2, 8, 11). Traditional approaches to imaging may overlook contributions from other lower extremity muscles and limit our understanding of how these muscles influence jump performance.

Advancements in machine learning technology have enabled the rapid measurement of muscle volumes from magnetic resonance imaging (MRI) scans, offering the opportunity to quantify various lower extremity muscle volumes. This technology has demonstrated utility in assessing associations between muscle volume and sprint and vertical jump performance in male basketball players and untrained individuals, as well as persistent muscle atrophy post-surgery

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and in elite athlete populations (12, 13). This study aimed to determine which combination of lower extremity muscle volumes was most strongly associated with CMJ performance across a large subset of high-level college athletes.

METHODS

This cross-sectional study included a cohort of National Collegiate Athletic Association (NCAA) Division 1 American football, basketball, ice hockey, and soccer athletes from a single institution. Before their competitive season, athletes completed CMJ testing on force plates and underwent bilateral lower extremity MRI scans. All athletes provided informed consent for the use of their CMJ and MRI data as part of their respective parent studies, each of which was approved by the University of Wisconsin–Madison Health Sciences IRB. Athletes were eligible for inclusion if they were cleared to participate in sport activities without limitation in practice and games at the time of data collection and had both CMJ and MRI data available and collected within four months of each other.

Jump Performance

Athletes performed bilateral maximum effort CMJs on portable, dual force platforms that measured vertical forces at a sampling rate of 800 Hz, with the software's internal filtering applied (Leonardo Mechanograph; Novotec Medical, Pforzheim, Germany). Athletes started with one foot on each force plate and hands placed on their hips. Athletes were instructed to "jump as high as possible" while maintaining their hands on their hips throughout the jump. A jump trial was considered successful if athletes landed with one foot on each force plate and maintained their balance. Athletes completed up to three successful trials. If a trial was unsuccessful, the athlete repeated the attempt, up to a maximum of five total jump trials per testing session.

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Quantifying Muscle Volume

Athletes from all sports underwent MRI of the lumbopelvic region and bilateral lower extremities using a 1.5T SIGNA Voyager Air Edition Mobile MRI (GE HealthCare, Chicago, IL, USA) or a 3-T scanner (SIGNA Premier [GE Healthcare]) equipped with a flexible surface coil. Imaging was performed continuously from the 12th thoracic vertebra to below the ankle joint in sets of 40 scans to capture the entire field of view. Axial multislice 2-dimensional fast spoiled gradient recalled echo (FSPGR) sequences were utilized, with parameters optimized to maximize the muscle signal and minimize the fat signal (repetition time/echo time: 839 ms/4.35 ms; field of view: 500 × 344 mm; matrix: 256 × 176; voxel size: 2 × 2 × 5 mm). In all instances, the acquired images were processed using automated machine-learning-based auto-segmentation algorithms to obtain 38 individual muscle volumes for each limb (Springbok Analytics, Charlottesville, VA), which has demonstrated repeatability comparable with interobserver variability (14, 15). Muscle volumes were then normalized to athletes' body mass and height ($mL \cdot kg^{-1} \cdot cm^{-1}$) (14).

Muscle and Jump Variable Selection

Jump height (cm) and peak power (W) were the primary variables of interest in the CMJ. Peak power during the concentric takeoff phase was assessed (i.e., the period of upward propulsion until foot-off), and jump height was calculated based on impulse momentum from the proprietary Leonardo Mechanograph software. Data from the trial with the highest jump height were extracted, and takeoff peak power from the same trial was normalized to body mass (W/kg) to allow for comparison across athletes. Given that the joint movements responsible for the CMJ primarily occur in the sagittal plane, we only included the 23 lower extremity muscles responsible for flexion and extension at the hip, knee, and ankle as candidate variables for

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analysis (Figure 1). Since the CMJ is a bilateral task, we summed the muscle volumes from both sides to obtain a single volume for each muscle to use for analysis.

Statistical Analyses

Descriptive statistics (mean ± standard deviation) were calculated for height, weight, jump height (cm), normalized peak power (W/kg), and normalized muscle volumes ($mL \cdot kg^{-1} \cdot cm^{-1}$). Multiple regression analyses with backwards selection were conducted to determine the combination of muscle volumes that best explained variance in jump height and peak power. Both sex (male, female) and sport (basketball, football, hockey, soccer) were forced into the model as potential covariates. The analysis began with a full model that included all 23 muscle volumes. The least contributing muscle was removed at each iteration based on the Bayesian Information Criterion (BIC), calculated to penalize additional candidate muscles and reduce overfitting. The process continued until no muscle removed from the model decreased the Bayesian Information Criterion (BIC) score by greater than two points (16). Effect estimates (β) and their 95% confidence intervals (CI) are reported, as well as the unadjusted and adjusted R^2 values.

Given the model findings, a post hoc F-test comparing nested linear models was run as a separate analysis to test the significance of the interaction between biceps femoris long head and vastus lateralis in the peak power model. Type III ANOVAs were performed on the final regression models to assess the overall effect of sport. Multicollinearity was assessed using the variance inflation factor (VIF) to identify candidate variables exceeding the threshold of five. All statistical analyses were performed in R (Version 4.4.1; R Core Team 2024), with the following packages: broom (v1.0.7), car (v3.1.2), corrplot (v0.92), dplyr (v1.1.4), ggcorrplot (v0.1.4.1), ggplot2 (v3.5.1), lm.beta (v1.7.2), MASS (v7.3.60.2).

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RESULTS

Two hundred and seven collegiate athletes were included in the study (Supplemental Table 1, Supplemental Digital Content, http://links.lww.com/MSS/D362), comprised of 164 males (height: 187.3 ± 7.6 cm, weight: 98.4 ± 20.0 kg) and 43 females (172.0 ± 8.1 cm, 69.5 ± 10.5 kg).

Following backward selection, the final model for CMJ jump height (Fig. 2; Supplemental Table 2, Supplemental Digital Content, http://links.lww.com/MSS/D362) explained 54% of the variance in jump height ($R^2 = 0.56$, adjusted $R^2 = 0.54$). The model included the vastus lateralis (estimate [95% CI] = 1.39 [0.91, 1.87], $\beta = 0.36$, $p < 0.01$) and rectus femoris (1.20 [-0.11, 2.51], $\beta = 0.11$, $p = 0.07$) muscle volumes after adjusting for sex and sport. A 1 $mL \cdot kg^{-1} \cdot cm^{-1}$ greater vastus lateralis muscle volume related to a 1.39 [0.91, 1.87] cm higher jump height. A Type III ANOVA indicated a significant main effect of sport ($F(3, 200) = 7.68$, $p < .001$). Marginal sport and sex specific regression coefficients were examined relative to basketball (reference sport) and female (reference sex), and both are adjusted for all other variables in the model. Hockey (6.63 [3.63, 9.63], $\beta = 0.27$, $p < 0.01$) and football (3.76 [1.26, 6.26], $\beta = 0.22$, $p < 0.01$) were associated with higher jump heights, whereas no difference was noted in soccer when compared to basketball players. On average, males also jumped higher than females (8.86 [6.52, 11.20], $\beta = 0.43$, $p < 0.01$).

The final model for peak power (Fig. 2; Supplemental Table 3, Supplemental Digital Content, http://links.lww.com/MSS/D362) explained 71% of the variance in peak power ($R^2 = 0.72$, adjusted $R^2 = 0.71$). The model included vastus lateralis (1.67 [1.20, 2.14], $\beta = 0.41$, $p < .01$), gluteus medius (1.16 [-0.04, 2.36], $\beta = 0.11$, $p = 0.06$), biceps femoris long head (-1.78, [-3.36, -0.20], $\beta = -0.11$, $p = 0.03$), adductor magnus (0.66 [0.12, 1.20], $\beta = 0.13$, $p = 0.02$), and

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soleus (1.54 [0.63, 2.44], $\beta = 0.16$, $p < 0.01$) muscle volumes after adjusting for sex and sport. A 1 $mL \cdot kg^{-1} \cdot cm^{-1}$ greater vastus lateralis muscle volume related to a 1.67 [1.20, 2.14] W/kg higher peak power. A Type III ANOVA indicated a significant main effect of sport ($F(3, 197) = 11.18$, $p < .001$). Specifically, football (4.49 [2.26, 6.72], $p < .01$) was associated with greater peak power compared to basketball players, whereas hockey or soccer, compared to basketball, was not statistically significant. Males also generated greater peak power than females (8.79 [6.75, 10.83], $p < .01$).

Multicollinearity analysis revealed that all muscle volumes in the final models had VIF values below the threshold of five.

DISCUSSION

This study used a dataset of over 200 NCAA Division 1 college athletes to determine the relationship between lower extremity muscle volumes and CMJ performance. After accounting for sex and sport, vastus lateralis muscle volume was identified to be a major contributor to both jump height and peak power, underscoring its key role in vertical jump performance. In addition to the vastus lateralis, which was the only muscle common to both regression models, the jump height model included the rectus femoris, while the peak power model retained the soleus, gluteus medius, adductor magnus, and biceps femoris long head. The differing muscle candidates in the two models likely reflect the distinct features of each CMJ metric: jump height reflects force over time (impulse, N·s) while peak power reflects the rapid generation of force (N·m/s) (17).

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Importance of the Quadriceps in Jump Performance

Quadricep muscles facilitate extension of the knee, providing the primary vertical propulsion for the lower extremities during jumping (8, 18). Prior studies assessing a select number of lower extremity muscles have shown relationships between quadriceps strength and thickness and jump performance (10, 19). In our analysis, vastus lateralis was strongly associated with both jump height and peak power when included amongst an expanded pool of lower extremity muscles. The vastus lateralis was highly correlated with the vastus medialis and vastus intermedius volumes (Supplemental Table 4, Supplemental Digital Content, http://links.lww.com/MSS/D362), suggesting it may be representative of overall quadriceps muscle volume. In addition, the vastus lateralis is the largest of the quadriceps muscles, making it particularly influential in vertical propulsion during jumping (20, 21). Given these findings, targeting hypertrophy of the vastus lateralis, and by extension the quadriceps, may be a promising strategy to enhance jump performance, although longitudinal studies are needed to establish a causal relationship.

Besides the vastus lateralis, both jump height and peak power regression models contained additional unique muscles, though their influences were less pronounced. In the jump height model, the rectus femoris was the only other muscle retained. Given that it crosses both the hip and knee and undergoes minimal length change during the CMJ, the rectus femoris may primarily transfer energy between joints rather than significantly contribute to vertical propulsion (22, 23). However, because it spans both joints, greater rectus femoris volume may allow for more effective force transmission from the hip to the knee during push-off and help transfer energy during rapid knee extension. Increased muscle volume may also reflect adaptations in muscle architecture (e.g., greater physiological cross-sectional area and stiffness) that enable the

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rectus femoris to sustain and transmit higher forces and enhance inter-joint energy transfer during the CMJ. In the peak power model, soleus, gluteus medius, adductor magnus, and biceps femoris long head were retained alongside the vastus lateralis. The soleus primarily contributes to vertical ground reaction force through ankle plantarflexion and indirectly assists knee extension through closed-chain mechanics, though it has less mechanical advantage at the knee than the vastus lateralis in vertical jump (24, 25). Although the gluteus medius and adductor magnus aid in hip extension, their activation patterns suggest their primary role may be stabilizing the hip joint, thereby optimizing force transfer through the lower extremity during the CMJ (25).

Conversely, our findings showed that smaller biceps femoris long head muscle volume was related to greater jump power. An interaction effect was observed between vastus lateralis and biceps femoris long head muscle volumes in our post hoc analysis (Supplemental Table 5, Supplemental Digital Content, http://links.lww.com/MSS/D362), demonstrating that a possible trade-off between muscle volume distribution and peak jump power may exist. This finding highlights the possibility that greater muscle size may not always be beneficial toward the task of interest, in this case the CMJ. As such, understanding how specific muscle size distributions impact lower extremity biomechanics and power output provides deeper insights into optimizing jump mechanics.

Sport Differences

On average, hockey players jumped the highest (Table 1), likely reflecting their sport-specific demands, which require consistent, powerful hip and knee extension for skating strides (26). Male hockey players also had notably larger normalized quadriceps volumes, particularly in the vastus lateralis and vastus medialis, which may also explain their greater jump heights

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(Supplemental Table 6, Supplemental Digital Content, http://links.lww.com/MSS/D362). Basketball players, despite frequently jumping during sport, demonstrated lower jump heights compared to hockey, football, and soccer athletes. Our findings also indicated that athletes with larger quadricep muscle volumes relative to body size generally produced greater peak power. This relationship was most pronounced in male hockey players.

Limitations

As this study involved high-level collegiate athletes at a single institution, the findings may not be fully generalizable to recreational, high school, or professional populations. Second, while our models identified muscle volumes significantly associated with CMJ performance, these muscles may not contribute similarly to other movements required in each specific sport. Third, given the cross-sectional study design, there is no direct evidence to support that training for hypertrophy in these identified muscle groups will result in better jump performance. Fourth, because this was an exploratory analysis, it is important to note that our use of multiple regression procedures (including backward selection and a post hoc interaction test) may have increased the risk of type I error. Lastly, the measurement of muscle volume provides unique insights into characterizing individual muscle contribution to CMJ, but is not a direct measure of muscle strength.

CONCLUSIONS

We identified that quadriceps muscle volume, particularly the vastus lateralis, was most strongly associated with CMJ height and peak power in elite athletes after accounting for sex and sport. While other muscles, such as the gluteus medius, rectus femoris, biceps femoris long head, adductor magnus, and soleus, were related to jump performance, their influence was less pronounced. These findings highlight the critical role of the quadriceps in vertical jump

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performance and suggest that strength and conditioning and rehabilitation programs may consider targeting quadriceps hypertrophy to improve vertical jump performance.

Acknowledgements

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FIGURE LEGENDS

Figure 1: Anterior and posterior views of the muscle volumes included in the backward regression models for jump height and peak power.

Figure 2: Significant muscle volume candidates of (a) jump height and (b) peak power. Solid lines represent regression fits; shaded regions indicate 95% confidence intervals.

SUPPLEMENTAL DIGITAL CONTENT

SDC 1: Supplemental Digital Content.docx

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Table 1: Athlete Demographics

Sex	Sport	n (%)	Height, cm*	Weight, kg*	Jump Height, cm*	Peak Power, W/kg*
Female	**Total**	43 (20.8%)	172.0 (8.1)	69.5 (10.5)	32.7 (5.1)	44.4 (5.4)
	Basketball	11 (5.3%)	177.1 (10.3)	76.2 (12.9)	32.3 (3.6)	44.0 (6.3)
	Hockey	10 (4.8%)	172.4 (7.6)	73.2 (10.6)	36.3 (6.0)	44.1 (5.9)
	Soccer	22 (10.6%)	169.3 (5.9)	64.4 (6.1)	31.2 (4.8)	44.8 (4.9)
Male	**Total**	164 (79.2%)	187.3 (7.6)	98.4 (20.0)	44.1 (7.4)	57.7 (7.4)
	Basketball	18 (8.7%)	194.5 (8.4)	93.4 (11.0)	38.9 (6.8)	55.2 (7.1)
	Hockey	18 (8.7%)	184.2 (5.2)	86.9 (5.4)	49.7 (6.0)	58.8 (4.6)
	Soccer	21 (10.1%)	180.9 (6.4)	77.8 (4.7)	44.0 (6.2)	54.2 (4.6)
	Football	107 (51.7%)	187.9 (6.8)	105.3 (20.7)	44.0 (7.4)	58.7 (8.0)
Total	**Total**	**207 (100.0%)**	**184.1 (9.9)**	**92.4 (21.9)**	**41.7 (8.4)**	**55.0 (8.9)**

*All values reported as mean (SD).

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Table 2: Final Multivariable Regression Models

Outcome Variable	Predictor Category	Predictor	Estimate β	95% CI	t-value	p-value
Jump Height (cm)	Sex	Male	8.86	[6.52, 11.20]	7.46	< .01
	Sport	Basketball	Reference			
		Football	3.76	[1.25, 6.26]	2.96	0.003
		Hockey	6.63	[3.63, 9.63]	4.36	< .001
		Soccer	1.42	[-1.32, 4.15]	1.02	0.31
	Muscle	Vastus Lateralis	1.39	[0.91, 1.86]	5.72	< .001
		Rectus Femoris	1.20	[-0.11, 2.51]	1.80	0.07
Peak Power (W/kg)	Sex	Male	8.79	[6.75, 10.83]	8.50	< .001
	Sport	Basketball	Reference			
		Football	4.49	[2.26, 6.72]	3.97	< .001
		Hockey	-0.53	[-3.15, 2.09]	-0.40	0.69
		Soccer	-1.97	[-4.56, 0.62]	-1.50	0.13
	Muscle	Gluteus Medius	1.16	[-0.04, 2.36]	1.91	0.06
		Vastus	1.67	[1.20, 2.14]	6.99	< .001

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Outcome Variable	Predictor Category	Predictor	Estimate β	95% CI	t-value	p-value
		Lateralis				
		Soleus	1.54	[0.63, 2.44]	3.36	< .001
		Biceps Femoris Long Head	-1.78	[-3.36, -0.20]	-2.23	0.03
		Adductor Magnus	0.66	[0.12, 1.20]	2.42	0.02

*Basketball was used as a reference for sport. Female was used as a reference for sex.

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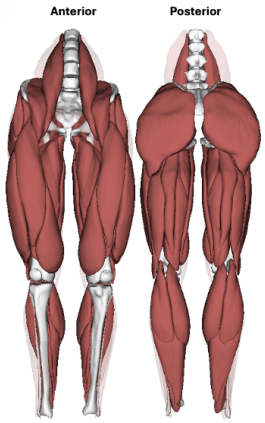
Figure 1



Muscle Group	Muscle
Hip Flexors	Psoas Major, Iliacus, Adductor Longus, Adductor Brevis
Hip Extensors	Gluteus Maximus, Gluteus Medius, Gluteus Minimus, Adductor Magnus
Knee Flexors	Biceps Femoris Long Head, Biceps Femoris Short Head, Semitendinosus, Semimembranosus, Gracilis
Knee Extensors	Rectus Femoris, Vastus Lateralis, Vastus Intermedius, Vastus Medialis
Ankle Plantarflexors	Gastrocnemius Medial Head, Gastrocnemius Lateral Head, Soleus, Posterior Tibialis, Flexor Digitorum Longus, Flexor Hallicus Longus

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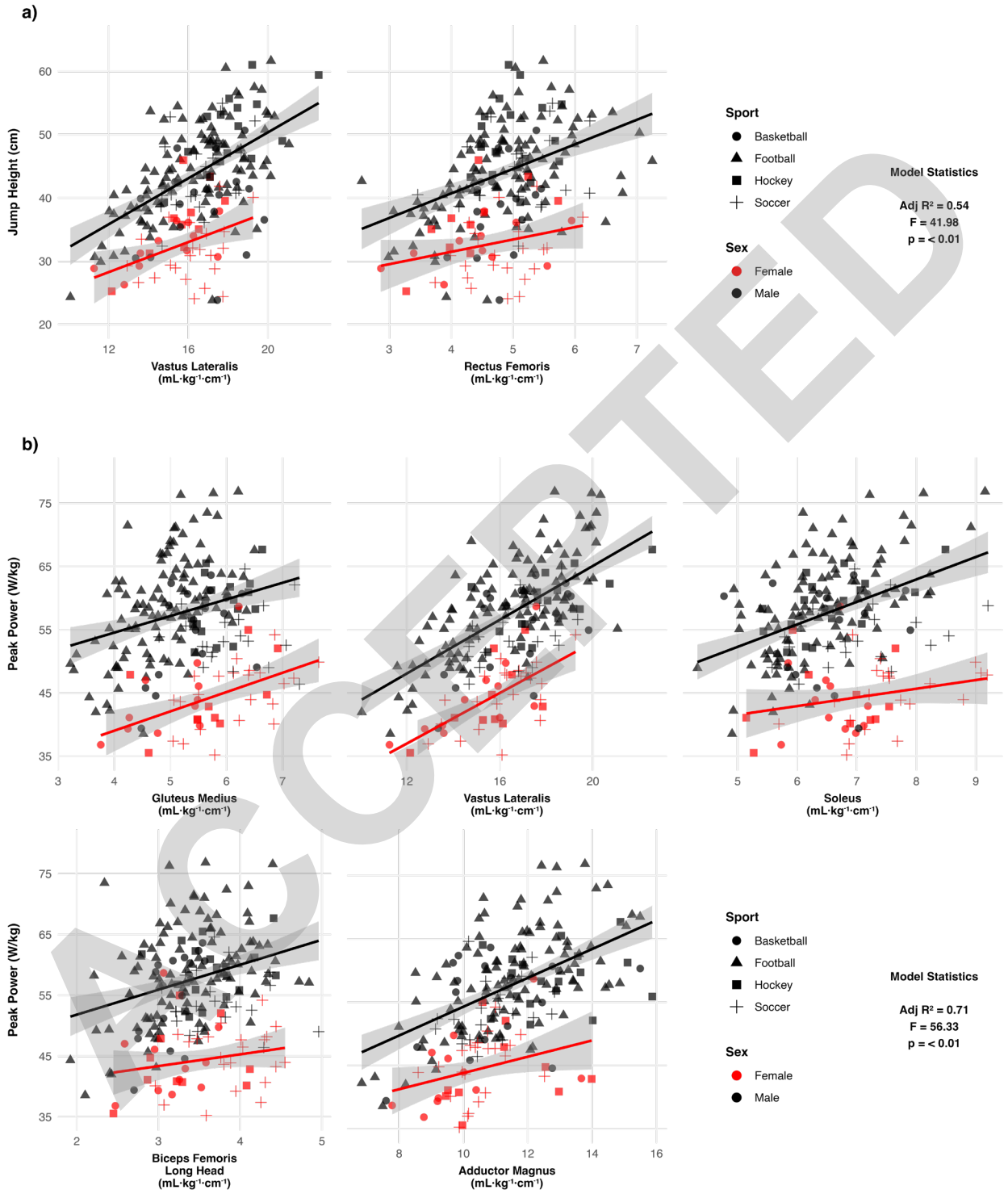
Figure 2



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